

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
AHP	Air High Pressure	Air (Instrument)	Flow Control	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) Pressure rating: PN16	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Ball	Compressed Air High Pressure generally refers to operating pressure being >= 10 Bar g
AHP	Air High Pressure	Air (Instrument)	Flow Control	Butterfly	ALL	Body: A216 Type WCB Disc: SS316 Shaft: SS 316 Seat: PTFE Seal: PTFE	WF, LG or F		Compressed Air flow regulation where applicable and where regulation is not facilitated by VFD air delivery equipment
AHP	Air High Pressure	Air (Instrument)	Isolation	Ball	50mm and below	Body: A351 Gr. CF8M Stem: A276 Type 316 or A479/A479M Type 316 Gland: A276 Type 316 or A351 Gr. CF8 Gland packing: PTFE Gasket: PTFE Ball Seat: Hypatite PTFE	F, S		Compressed Air High Pressure generally refers to operating pressure being >= 10 Bar g
AHP	Air High Pressure	Air (Instrument)	Isolation	Ball	Above 100mm	Body: A351 Gr. CF8M Stem: A276 Type 316 or A479/A479M Type 316 Gland: A276 Type 316 or A351 Gr. CF8 Gland packing: PTFE Gasket: PTFE Ball Seat: Hypatite PTFE	F		Compressed Air High Pressure generally refers to operating pressure being >= 10 Bar g
AHP	Air High Pressure	Air (Instrument)	Non-return	Check-Dual Flap	50mm and below	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM / SS316 Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		Compressed Air High Pressure generally refers to operating pressure being >= 10 Bar g
AHP	Air High Pressure	Air (Instrument)	Pressure Reducing	Pressure Reducing	50mm and below	Type: Spring control Body: SS316 Pressure adjustment spring: SS316 Diaphragm: SS316 Adjustment screw: A4-80 Valve seat: SS 431	F, S		Compressed Air High Pressure generally refers to operating pressure being >= 10 Bar g
ALP	Air Low Pressure	Aeration	Flow Control	Jet	ALL	Body: SS316 Disc: SS316 Shaft: SS 316 Seat: PTFE Seal: PTFE	WF, LG or F	Diaphragm	Compressed Air flow regulation where applicable and where regulation is not facilitated by VFD air delivery equipment
ALP	Air Low Pressure	Aeration	Isolation	Butterfly	ALL	Design standard: BS EN 593, AWWA C504-15 Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Ball	Low Pressure generally refers to operating pressure being < 10 Bar g
ALP	Air Low Pressure	Aeration	Non-return	Check-Dual Flap	Above 50 mm	Design standard: API 594 / ISO 5758 / DIN 3202 K3 Body: DI with Epoxy coating / SS316 Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		Corrosive applications Low Pressure generally refers to operating pressure being < 10 Bar g

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ALP	Air Low Pressure	Air Scouring	Non-return	Check-Dual Flap	Above 50 mm	Design standard: API 594 / ISO 5758 / DIN 3202 K3 Body: DI with Epoxy coating / SS316 Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		Low Pressure generally refers to operating pressure being < 10 Bar g
ALP	Air Low Pressure	Aeration	Pressure Reducing	Pressure Reducing	50mm and below	Type: Spring control Body: SS316 Pressure adjustment spring: SS316 Diaphragm: SS316 Adjustment screw: A4-80 Valve seat: SS 431	F		Low Pressure generally refers to operating pressure being < 10 Bar g
ASC	Antiscalant	Antiscalant	Isolation	Ball	All	Body: CPVC Stem: CPVC Ball: CPVC Seat: PTFE O-ring: FKM or EPDM	F / Union Coupling	Butterfly	
ASC	Antiscalant	Antiscalant	Non-return	Duckbill	750 mm and below	Body: EPDM	F		For end of overflow pipe
ASC	Antiscalant	Antiscalant	Non-return	Ball Check	All	Body: PP Ball: PP Seat: FKM/PTFE O-Ring: FKM/PTFE	F	Swing Check	
ASC	Antiscalant	Antiscalant	Pressure loading	Pressure Loading	All	Body: PP Piston: PP Diaphragm: PTFE Seal: EPDM	F / Union Coupling		
ASC	Antiscalant	Antiscalant	Safety pressure relief	Pressure Relief	All	Body: PP Piston: PP Diaphragm: PTFE Seal: EPDM	F / Union Coupling		
BL	BTF Liquor	Biotrickling Filter Water Supply/Recirculation	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating / CPVC Disc: SS 316 (1.4408 CF8M) / CPVC or PP-H Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) / CPVC	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Globe, Ball	
BL	BTF Liquor	Biotrickling Filter Water Supply/Recirculation	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating / CPVC Disc: SS 316 (1.4408 CF8M) / CPVC or PP-H Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) / CPVC	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Ball	
BL	BTF Liquor	Biotrickling Filter Water Supply/Recirculation	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F		
CDG	Carbon Dioxide (Gas)	Carbon Dioxide	Isolation	Butterfly	50 mm and above	Body: Cast stainless steel 1.4409 Disc: SS 1.4404 (i.e. SS316) Stem: SS 1.4404 (i.e. SS316) Class: as required according to design pressure rating.	LG		
CDG	Carbon Dioxide (Gas)	Carbon Dioxide	Non-return	Swing Check	50 mm and above	Body: SS316 casting - CF8M Disc: SS316 casting - CF8M Hinge: SS316 Class: as required according to design pressure rating	WF		

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Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
CIT	Citric Acid	Citric Acid	Isolation	Ball	50 mm and above	Body: CPVC Stem: CPVC Ball: CPVC Seat: PTFE O-ring: FKM or EPDM	F / Union Coupling	Butterfly	
CIT	Citric Acid	Citric Acid	Non-return	Ball Check	All	Body: SS 316 Casting CF8M Ball: Aluminium vulcanised with NBR rubber / Grey Cast Iron vulcanised with NBR rubber / Glass filled PTFE / Ceramic Seat: NBR / EPDM / PTFE	F	Swing Check	
CIT	Citric Acid	Citric Acid	Pressure loading	Pressure Loading	All	Body: PP Piston: PP Diaphragm: PTFE Seal: EPDM	F / Union Coupling		
CIT	Citric Acid	Citric Acid	Safety pressure relief	Pressure Relief	All	Body: PP Piston: PP Diaphragm: PTFE Seal: EPDM	F / Union Coupling		
DC	Dewatering Centrate	Dewatering Centrate	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
DC	Dewatering Centrate	Dewatering Centrate	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F		
DCW	Dewatering Centrate (WAS Pre-dewatering)	Dewatering Centrate (WAS Pre-dewatering)	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
DCW	Dewatering Centrate (WAS Pre-dewatering)	Dewatering Centrate (WAS Pre-dewatering)	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F		
DS	Digested Sludge	Digested Sludge	Draw-Off	Telescopic	All	Frame: SS316L Cover Slide: SS316L Yoke: SS316L Stem:SS316L Gasket: EPDM/Neoprene rubber	F		

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DS	Digested Sludge	Digested Sludge	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
DS	Digested Sludge	Digested Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F	Swing Check	
DUW	Domestic Used Water	Raw sewage	Flow Control	Eccentric Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) of pinless connection to disc	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
DUW	Domestic Used Water	Raw sewage	Isolation	Knife gate	600 mm and below	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
DUW	Domestic Used Water	Raw sewage	Isolation	Knife gate	650 mm and above	As per General Specifications	F		
DUW	Domestic Used Water	Raw sewage	Non-return	Ball Valve designed to also function as check (refer PDRR)	All	Ball: Ductile ironn ASTM A536 / Cast iron ASTM A48 Shaft: Type 316 SS / Type 630 SS Seat (Body): Monel Alloy UNS: N04060 Seat (Rotor): ASTM A351 Grade CF8M O-rings: Buna-N	F		Corrosion resistant coating where applicable
DWAS	Dewatered Waste Activated Sludge	Dewatered Waste Activated Sludge	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		

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Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
FA	Foul Air	Exhaust Air	Flow control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: FRP (eg. Durco 62) / CPVC Disc: PFA Lined DI / CPVC or PP-H Liner/Seat: PTFE Shaft/Stem/Spindle: PTFE coated ASTM A193 / CF8M / CPVC	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Stainless Steel for high temperature (>60°C and THP process gas)	
FA	Foul Air	Exhaust Air	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: FRP (eg. Durco 62) / CPVC Disc: PFA Lined DI / CPVC or PP-H Liner/Seat: PTFE Shaft/Stem/Spindle: PTFE coated ASTM A193 / CF8M / CPVC	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Stainless Steel for high temperature (>60°C and THP process gas)	
FA	Foul Air	Exhaust Air	Non-return	Swing Check	All	Design standard: API 594 / DIN 3202 K3 Body: FRP Plate: FRP Seat: PTFE/Viton	WF	Stainless Steel for high temperature (>60°C and THP process gas)	
FC	Ferric Chloride	Ferric Chloride	Isolation	Butterfly	50 mm and above	Body: CPVC Disc: CPVC Liner/Seat: EPDM Shaft/Stem/Spindle: SS316 fully sealed off from fluid	LG		
FC	Ferric Chloride	Ferric Chloride	Non-return	Duckbill	750 mm and below	Body: EPDM	F		For end of storage tank overflow pipe
FC	Ferric Chloride	Ferric Chloride	Non-return	Ball Check	All	Body: PP Ball: PP Seat: FKM/PTFE O-Ring: FKM/PTFE	F	Swing Check	
FW	Food Waste	Food Waste	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
FW	Food Waste	Food Waste	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F	Swing Check	
G2D	Plant Water (DLM MF)	Plant Water	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	<80mm NB Ball Valve	

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Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
G2D	Plant Water (DLM MF)	Plant Water	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	<80mm NB Ball Valve	
G2D	Plant Water (DLM MF)	Plant Water	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
G2D	Plant Water (DLM MF)	Plant Water	Pressure Reducing	Pressure Reducing	50mm and below	Type: Spring control Body: SS316 Pressure adjustment spring: SS316 Diaphragm: SS316 Adjustment screw: A4-80 Valve seat: SS 431	F		
G2HP	Plant Water High Pressure	Plant Water	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) Pressure rating: PN16	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
G2HP	Plant Water High Pressure	Plant Water	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) Pressure rating: PN16	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Gate	
G2HP	Plant Water High Pressure	Plant Water	Isolation	Ball	All	Body: A216 Gr. WCB Stem: A276 Type 316 Gland: A351 Gr. CF8M Gland: A351 Gr. CF8 Gasket: PTFE Ball Seat: Hypatite PTFE	F, S		
G2HP	Plant Water High Pressure	Plant Water	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
G2I	Plant Water (ILM MF)	Plant Water	Air release/Vacuum relief	Combination Air	All	Body: Stainless Steel ASTM A351 Grade CF8M Cover: Stainless Steel ASTM A351 Grade CF8M Trim: Type 316 Stainless Steel Float: Stainless steel ASTM A240 Coating: Non stick fusion bonded epoxy	F		

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G2I	Plant Water (ILM MF)	Plant Water	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	<80mm NB Ball Valve	
G2I	Plant Water (ILM MF)	Plant Water	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	<80mm NB Ball Valve	
G2I	Plant Water (ILM MF)	Plant Water	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
GB	Gas (Biogas)	Gas (Biogas)	Flow Control	Butterfly	50 mm and above	Body: Cast stainless steel 1.4409 Disc: SS 1.4404 (i.e. SS316) Stem: SS 1.4404 (i.e. SS316) Class: as required according to design pressure rating.	F, LG		
GB	Gas (Biogas)	Gas (Biogas)	Isolation	Butterfly	50 mm and above	Body: Cast stainless steel 1.4409 Disc: SS 1.4404 (i.e. SS316) Stem: SS 1.4404 (i.e. SS316) Class: as required according to design pressure rating.	F, LG		
GB	Gas (Biogas)	Gas (Biogas)	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: SS316 Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
GR	Grit	Grit Slurry	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		

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Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
GW	Greasy Waste	Greasy Waste	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
GW	Greasy Waste	Greasy Waste	Non-return	Ball Check	All	Body: Up to DN 200mm SS 316 Casting CF8M / Above DN 200mm DI Ball: Aluminium vulcanised with NBR rubber / Grey Cast Iron vulcanised with NBR rubber / Glass filled PTFE / Ceramic Seat: NBR / EPDM / PTFE	F	Swing Check	
HS	Hydrolysed Sludge	Hydrolysed Sludge	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		To be approved by THP specialist supplier.
HS	Hydrolysed Sludge	Hydrolysed Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F	Swing Check	To be approved by THP specialist supplier.
HWR	Hot Water (Return)	Hot Water (Return)	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
HWR	Hot Water (Return)	Hot Water (Return)	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
HWR	Hot Water (Return)	Hot Water (Return)	Isolation	Ball	All	Body: A216 Gr. WCB Stem: A276 Type 316 Gland: A351 Gr. CF8M Gland: A351 Gr. CF8 Gasket: PTFE Ball Seat: Hypatite PTFE	F, S		

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Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
HWR	Hot Water (Return)	Hot Water (Return)	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
IUW	Industrial Used Water	Industrial Used Water	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) of pinless connecction to disc	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
IUW	Industrial Used Water	Raw sewage	Flow Control	Plug	All	Plug: ASTM A351 CF8M Gasket: PTFE Sleeve: PTFE Bonnet: ASTM A351 CF8M Cover: ASTM A351 CF8M	F		Corrosion resistant coating where applicable

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IUW	Industrial Used Water	Industrial Used Water	Isolation	Knife Gate	600 mm and below	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
IUW	Industrial Used Water	Raw sewage	Isolation	Knife gate	650 mm and above	As per General Specifications	F		
IUW	Industrial Used Water	Raw sewage	Non-return	Ball Valve designed to also function as check (refer PDRR)	All	Ball: Ductile iron ASTM A536 / Cast iron ASTM A48 Shaft: Type 316 SS / Type 630 SS Seat (Body): Monel Alloy UNS: N04060 Seat (Rotor): ASTM A351 Grade CF8M O-rings: Buna-N	F		Corrosion resistant coating where applicable
IW	Industrial Water	Industrial Water	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) of pinless connection to disc	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
IW	Industrial Water	Industrial Water	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) of pinless connection to disc	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
IW	Industrial Water	Industrial Water	Non-return	Nozzle Check	All	Body: ASTM A351 CF8M Seat ring: ASTM A351 CF8M Valve disc: AISI 316 Spring: AISI 316 Guide bush: AISI 316 Flow diffuser: ASTM A351 CF8M	F		
LS	Lime Solution	Lime	Isolation	Diaphragm	All	Body: Cast Iron Diaphragm: EPDM / FPM Stem: SS316 Seal: EPDM / FPM	F	Pinch Valve to be used for lime slurry	
LS	Lime Solution	Lime	Non-return	Swing Check	50 mm and above	Body: CPVC Disc: CPVC Spindle: CPVC Gasket: EPDM	WF		
LSL	Lime Sludge	Lime	Isolation	Pinch	All	Body: Cast Iron Flexible sleeve: EPDM or NBR Stem and Pinch bars: Carbon Steel	F		
LSL	Lime Sludge	Lime	Non-return	Swing Check	50 mm and above	Body: CPVC Disc: CPVC Spindle: CPVC Gasket: EPDM	WF		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
MF	Membrane Filtrate	Filtrate	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
BWS	Backwash Supply	Backpulse	Isolation	Knife gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS316 Stem: SS316 Seat: EPDM	F		
BWS	Backwash Supply	Backpulse	Isolation	Knife gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS316 Stem: SS316 Seat: Brass or bronze or SS	F		
BWS	Backwash Supply	Backpulse	Isolation	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
MF	Membrane Filtrate	Filtrate	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Ball	
BWS	Backwash Supply	Backpulse	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: SS316 Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
ML	Mixed Liquor	Mixed Liquor	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F		

Appendix P4C: Valve Selection Guideline

Instruction:

1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
NHS	Ammonium Sulphate	Ammonium Sulphate	Isolation	Ball	50 mm and above	Body: CPVC Stem: CPVC Ball: CPVC Seat: PTFE O-ring: FKM or EPDM	F / Union Coupling	Butterfly	
NHS	Ammonium Sulphate	Ammonium Sulphate	Non-return	Ball Check	All	Body: SS 316 Casting CF8M Ball: Aluminium vulcanised with NBR rubber / Grey Cast Iron vulcanised with NBR rubber / Glass filled PTFE / Ceramic Seat: NBR / EPDM / PTFE	F	Swing Check	
NHS	Ammonium Sulphate	Ammonium Sulphate	Pressure loading	Pressure Loading	All	Body: PP Piston: PP Diaphragm: PTFE Seal: EPDM	F / Union Coupling		
NHS	Ammonium Sulphate	Ammonium Sulphate	Safety pressure relief	Pressure Relief	All	Body: PP Piston: PP Diaphragm: PTFE Seal: EPDM	F / Union Coupling		
NW	NEWater	NEWater	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) of pinless connecction to disc	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
NW	NEWater	NEWater	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with Rilsan or Halar coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) of pinless connecction to disc	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
NW	NEWater	NEWater	Non-return	Nozzle Check	All	Body: ASTM A351 CF8M Seat ring: ASTM A351 CF8M Valve disc: AISI 316 Spring: AISI 316 Guide bush: AISI 316 Flow diffuser: ASTM A351 CF8M	F		
PD	Plant Drain	Plant Drain	Air release/Vacuum relief	Combination Air	All	Body and Cover: Ductile iron ASTM A536 Float: Stainless steel ASTM A240 Stem: Stainless steel series T300 Seat: Nitrile Rubber Exterior paint: Universal metal primer	F		
PD	Plant Drain	Plant Drain	Isolation	Knife Gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM	F		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
PD	Plant Drain	Plant Drain	Isolation	Knife Gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F		
PD	Plant Drain	Plant Drain	Isolation	Ball	All	Body: A216 Gr. WCB with corrosion resistant epoxy coating Stem: A276 Type 316 Gland: A351 Gr. CF8M Gland: A351 Gr. CF8 Gasket: PTFE Ball Seat: Hypatite PTFE	F, S	Butterfly (Only to be used if there is a low risk of blockage, for example on processes downstream of fine screens)	
PD	Plant Drain	Plant Drain	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F		
PE	Primary Effluent	Primary effluent	Flow Control	Eccentric Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
PE	Primary Effluent	Primary effluent	Isolation	Knife Gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM	F	<80mm NB Ball Valve	
PE	Primary Effluent	Primary effluent	Isolation	Knife Gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F	Plug Valve	
PE	Primary Effluent	Primary effluent	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F	Ball Check	
PLE	Plant Effluent	Plant Water	Flow Control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) Pressure rating: PN16	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
PLE	Plant Effluent	Plant effluent	Flow Control	Eccentric Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
PLE	Plant Effluent	Plant effluent	Isolation	Knife gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM	F		
PLE	Plant Effluent	Plant effluent	Isolation	Knife gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F		
PLE	Plant Effluent	Plant Water	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Gate	
PLE	Plant Effluent	Plant Water	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
PLE	Plant Effluent	Plant effluent	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F		
PSD	Primary Sludge Domestic	Primary Sludge	Flow Control	Eccentric Plug	All	Body: Cast Iron or Ductile Iron Plug: Cast Iron or Ductile Iron (with NBR encapsulation) Gasket: PTFE Seals: NBR	F	Ball	
PSD	Primary Sludge Domestic	Primary sludge	Isolation	Knife gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM	F		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
PSD	Primary Sludge Domestic	Primary sludge	Isolation	Knife gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F		
PSD	Primary Sludge Domestic	Primary Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F	Swing Check / Ball Check	
PSI	Primary Sludge Industrial	Primary Sludge	Flow Control	Eccentric Plug	All	Body: Cast Iron or Ductile Iron Plug: Cast Iron or Ductile Iron (with NBR encapsulation) Gasket: PTFE Seals: NBR	F	Ball	
PSI	Primary Sludge Industrial	Primary sludge	Isolation	Knife gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM	F		
PSI	Primary Sludge Industrial	Primary sludge	Isolation	Knife gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F		
PSI	Primary Sludge Industrial	Primary Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F		
PSI	Primary Sludge Industrial	Primary Sludge	Pressure relief	Safety Relief Valve	All	Body: SA 216 WCB Bonnet: SA 216 WCB Nozzle: 316L Adjusting ring: 316L: Hardened stainless steel Spring: High temperature alloy steel Lock screw: 316L	F		
PSM	Primary Scum	Scum	Air release/Vacuum relief	Combination Air	All	Body and Cover: Ductile iron ASTM A536 Float: Stainless steel ASTM A240 Stem: Stainless steel series T300 Seat: Nitrile Rubber Exterior paint: Universal metal primer	F		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
PSM	Primary Scum	Scum	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
PSM	Primary Scum	Scum	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F	Ball check	
RAS	Return Activated Sludge	Return Activated Sludge	Flow Control	Eccentric Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
RAS	Return Activated Sludge	Return Activated Sludge	Isolation	Knife gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F		
RAS	Return Activated Sludge	Return Activated Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F		
ROC	RO Concentrate	RO Concentrate	Flow Control	Globe	ALL	Globe valve shall comply with the General Specifications	T for 50mm or less; S for sizes larger than 50mm		
ROC	RO Concentrate	RO Concentrate	Isolation	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
ROC	RO Concentrate	RO Concentrate	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
ROF	RO Feed	RO Feed	Isolation	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
ROF	RO Feed	RO Feed	Non-return	Nozzle Check	All	Body: ASTM A351 CF8M Seat ring: ASTM A351 CF8M Valve disc: AISI 316 Spring: AISI 316 Guide bush: AISI 316 Flow diffuser: ASTM A351 CF8M	F		
ROP	RO Permeate	RO Permeate	Isolation	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr)	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
ROP	RO Permeate	RO Permeate	Non-return	Check-Dual Flap	All	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		
S	Sample	Sample (Water)	Isolation	Ball	All	Body: A351 Gr. CF8M Stem: A276 Type 316 Ball Seat: A351 Gr. CF8M or A276 Type 316 Gasket: PTFE Spring and pin: A313 and A276 Type 316	S, F	Plug	PCV ball valves can be used wherever the sample pipework is PVC
SA	Sulphuric Acid	Sulphuric Acid	Isolation	Ball	15 to 100 mm	Body: PVDF Ball: PVDF Stem: PVDF Seat: PTFE O-ring: FKM	F, Union coupling		For the discharge isolation valve and drain valve of 98 % sulphuric acid storage tank
SA	Sulphuric Acid	Sulphuric Acid	Non-return	Duckbill	750 mm and below	Body: Viton	F		For end of storage tank overflow pipe
SBS	Sodium Bisulphite	Sodium Bisulphite	Flow Control	Diaphragm	All	Body: CPVC Diaphragm: EPDM / FPM Stem: SS316 Seal: EPDM / FPM	F		
SBS	Sodium Bisulphite	Sodium Bisulphite	Isolation	Vented Ball	All	Body: CPVC Stem: CPVC Ball: CPVC Seat: PTFE O-ring: FKM or EPDM	F / Union Coupling		
SBS	Sodium Bisulphite	Sodium Bisulphite	Non-return	Duckbill	750 mm and below	Body: Viton	F		For end of storage tank overflow pipe
SBS	Sodium Bisulphite	Sodium Bisulphite	Non-return	Ball Check	All	Body: PP Ball: PP Seat: FKM/PTFE O-Ring: FKM/PTFE	F / Union Coupling		
SBS	Sodium Bisulphite	Sodium Bisulphite	Pressure loading	Pressure Loading	All	Body: CPVC Piston: CPVC Diaphragm: PTFE Seal: Viton	F / Union Coupling		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
SBS	Sodium Bisulphite	Sodium Bisulphite	Safety pressure relief	Pressure Relief	All	Body: CPVC Piston: CPVC Diaphragm: PTFE Seal: Viton	F / Union Coupling		
SC	Condensate	Condensate	Isolation	Ball	All	Body: CPVC Stem: CPVC Ball: CPVC Seat: PTFE O-ring: FKM or EPDM	F / Union Coupling	Stainless Steel for high temperature (>60°C)	
SC	Condensate	Condensate	Non-return	Ball Check	All	Body: PP Ball: PP Seat: FKM/PTFE O-Ring: FKM/PTFE	F / Union Coupling	Stainless Steel for high temperature (>60°C)	
SHYD	Sodium Hydroxide	Sodium Hydroxide	Isolation	Ball	50 mm and above	Body: CPVC Stem: CPVC Ball: CPVC Seat: PTFE O-ring: FKM or EPDM	F / Union Coupling	Butterfly	
SHYD	Sodium Hydroxide	Sodium Hydroxide	Non-return	Duckbill	750 mm and below	Body: Natural rubber	F		For end of storage tank overflow pipe
SHYD	Sodium Hydroxide	Sodium Hydroxide	Non-return	Ball Check	All	Body: PP Ball: PP Seat: FKM/PTFE O-Ring: FKM/PTFE	F / Union Coupling		
SHYD	Sodium Hydroxide	Sodium Hydroxide	Pressure loading	Pressure Loading	All	Body: CPVC Piston: CPVC Diaphragm: PTFE Seal: Viton	F / Union Coupling		
SHYD	Sodium Hydroxide	Sodium Hydroxide	Safety pressure relief	Pressure Relief	All	Body: CPVC Piston: CPVC Diaphragm: PTFE Seal: Viton	F / Union Coupling		
SHYP	Sodium Hypochlorite	Sodium Hypochlorite	Flow Control	Diaphragm	All	Body: CPVC Diaphragm: EPDM / FPM Stem: SS316 Seal: EPDM / FPM	F		
SHYP	Sodium Hypochlorite	Sodium Hypochlorite	Isolation	Vented Ball	All	Body: CPVC Stem: CPVC Ball: CPVC Seat: PTFE O-ring: FKM or EPDM	F / Union Coupling		
SHYP	Sodium Hypochlorite	Sodium Hypochlorite	Non-return	Duckbill	750 mm and below	Body: Viton	F		For end of storage tank overflow pipe
SHYP	Sodium Hypochlorite	Sodium Hypochlorite	Non-return	Ball Check	All	Body: PP Ball: PP Seat: FKM/PTFE O-Ring: FKM/PTFE	F / Union Coupling		
SHYP	Sodium Hypochlorite	Sodium Hypochlorite	Pressure loading	Pressure Loading	All	Body: CPVC Piston: CPVC Diaphragm: PTFE Seal: Viton	F / Union Coupling		
SHYP	Sodium Hypochlorite	Sodium Hypochlorite	Safety pressure relief	Pressure Relief	All	Body: CPVC Piston: CPVC Diaphragm: PTFE Seal: Viton	F / Union Coupling		

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
SSM	Secondary Scum	Scum	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
SSM	Secondary Scum	Scum	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F	Ball Check	
STHP	Steam High Pressure	Steam High Pressure	Flow control	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) Pressure rating: PN16	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Ball or Globe	High Pressure generally refers to operating pressure being >= 10 Bar g
STHP	Steam High Pressure	Steam High Pressure	Isolation	Ball	50mm and below	Body: A216 Gr. WCB or Ductile Iron Stem: A276 Type 316 or A479/A479M Type 316 Gland: A276 Type 316 or A351 Gr. CF8 Gland packing: PTFE Gasket: PTFE Ball Seat: Hypatite PTFE	F, S		High Pressure generally refers to operating pressure being >= 10 Bar g
STHP	Steam High Pressure	Steam High Pressure	Isolation	Ball	Above 100mm	Body: A216 Gr. WCB or Ductile Iron Stem: A276 Type 316 or A479/A479M Type 316 Gland: A276 Type 316 or A351 Gr. CF8 Gland packing: PTFE Gasket: PTFE Ball Seat: Hypatite PTFE	F		High Pressure generally refers to operating pressure being >= 10 Bar g
STHP	Steam High Pressure	Steam High Pressure	Non-return	Check-Dual Flap	50mm and below	Design standard: API 594 / DIN 3202 K3 Body: DI lined with EPDM / SS316 Plates: SS316 Shaft/Stem/Spindle: SS316 Spring: SS316 Liner: EPDM	WF or LG		High Pressure generally refers to operating pressure being >= 10 Bar g
TA	Treated Air	Exhaust Air	Flow control	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: FRP (eg. Durco 62) / CPVC Disc: PFA Lined DI / CPVC or PP-H Liner/Seat: PTFE Shaft/Stem/Spindle: PTFE coated ASTM A193 / CF8M / CPVC	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
TA	Treated Air	Exhaust Air	Isolation	Butterfly	All	Short body flange face to face dimens. as per BS EN 558 Body: FRP (eg. Durco 62) / CPVC Disc: PFA Lined DI / CPVC or PP-H Liner/Seat: PTFE Shaft/Stem/Spindle: PTFE coated ASTM A193 / CF8M / CPVC	LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm	Knife Gate	

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
TA	Treated Air	Exhaust Air	Non-return	Swing Check	All	Design standard: API 594 / DIN 3202 K3 Body: FRP Plate: FRP Seat: PTFE/Viton	WF		
TCD	Thickening Centrate Domestic	Thickening Centrate Recycle	Flow Control	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) Pressure rating: PN16	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
TCD	Thickening Centrate Domestic	Thickening Centrate Recycle	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
TCD	Thickening Centrate Domestic	Thickening Centrate Recycle	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F		
TCI	Thickening Centrate Industrial	Thickening Centrate Recycle	Flow Control	Butterfly	ALL	Short body flange face to face dimens. as per BS EN 558 Body: DI with epoxy coating Disc: SS 316 (1.4408 CF8M) Liner/Seat: EPDM (replaceable) Shaft/Stem/Spindle: SS 1.4029 (13 % Cr) Pressure rating: PN16	WF or LG or Semi-Lug for up to and including 200 mm; F or LG or Semi-Lug for sizes larger than 200 mm		
TCI	Thickening Centrate Industrial	Thickening Centrate Recycle	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
TCI	Thickening Centrate Industrial	Thickening Centrate Recycle	Non-return	Swing Check	All	Body: PFA lined ductile iron Disc: PFA lined ductile iron Hinge: PFA coated SS316	WF, LG or F		
TPS	Thickened Primary Sludge	Thickened Primary Sludge	Flow Control	Eccentric Plug	All	Body: Cast Iron or Ductile Iron Plug: Cast Iron or Ductile Iron (with NBR encapsulation) Gasket: PTFE Seals: NBR	F	Ball	

Appendix P4C: Valve Selection Guideline

Instruction:
1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
TPS	Thickened Primary Sludge	Thickened Primary Sludge	Isolation	Knife gate	All	Short body flange face to face dimens. as per BS EN 558 Type: To be resilient seated for size 300 mm and below but metal seated for size above 300 mm Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM for resilient seated type but brass, bronze or SS for metal seated type	F		
TPS	Thickened Primary Sludge	Thickened Primary Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F	Swing Check	
WASD	Waste Activated Sludge Domestic	Waste Activated Sludge	Flow Control	Eccentric Plug	All	Body: Cast Iron or Ductile Iron Plug: Cast Iron or Ductile Iron (with NBR encapsulation) Gasket: PTFE Seals: NBR	F	Ball	
WASD	Waste Activated Sludge Domestic	Waste Activated Sludge	Isolation	Knife gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM	F		
WASD	Waste Activated Sludge Domestic	Waste Activated Sludge	Isolation	Knife gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F		
WASD	Waste Activated Sludge Domestic	Waste Activated Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F		
WASI	Waste Activated Sludge Industrial	Waste Activated Sludge	Air release/Vacuum relief	Combination Air	All	Body and Cover: Ductile iron ASTM A536 Float: Stainless steel ASTM A240 Stem: Stainless steel series T300 Seat: Nitrile Rubber Exterior paint: Universal metal primer	F		
WASI	Waste Activated Sludge Industrial	Waste Activated Sludge	Flow Control	Eccentric Plug	All	Body: Cast Iron or Ductile Iron Plug: Cast Iron or Ductile Iron (with NBR encapsulation) Gasket: PTFE Seals: NBR	F	Ball	

Appendix P4C: Valve Selection Guideline

Instruction:

1. This Table shall serve as a guide to the Tenderers. Final valve selection shall be approved by the S.O. before procurement and installation.

Fluid Code	Process Fluid	Service	Duty	Type ¹	Size Range	Description ²	Connection ³	Alternative ⁴	Remarks
WASI	Waste Activated Sludge Industrial	Waste Activated Sludge	Isolation	Knife gate	300mm and below	Short body flange face to face dimens. as per BS EN 558 Type: Resilient seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: EPDM	F		
WASI	Waste Activated Sludge Industrial	Waste Activated Sludge	Isolation	Knife gate	Above 300mm	Short body flange face to face dimens. as per BS EN 558 Type: Metal seated Body: DI with corrosion resistant epoxy coating or SS Bonnet: DI with corrosion resistant epoxy coating or SS Body/Bonnet seal: EPDM Gate/Disc: SS 316 Stem: SS316 Seat: Brass or bronze or SS	F		
WASI	Waste Activated Sludge Industrial	Waste Activated Sludge	Non-return	Swing Check with counter-weight	All	Body: Ductile iron to BS EN 1563 Bonnet: Ductile iron to BS EN 1563 Lever: Ductile iron to BS EN 1563 Weight: Ductile iron to BS EN 1563 O-ring: EPDM Hinge: Stainless Steel SS316 Shaft: Stainless Steel SS316 Disc: EPDM Retainer washer: Stainless Steel SS316	F		

Notes:

1. Any valve actuators shall comply with the conditions of the general specification unless otherwise specified.
2. Where required, refer to Appendix P4A.4: Pipe Material Seleccction Guideline for abbreviations of materials.
3. F - Flanged of same PN rating as the pipe
WF - Wafer
LG - Lugged
S - Screwed (Threaded)
4. Shall be subjected to S.O. approval. Full valve details shall be submitted by Contractor.
5. Butterfly valves with discs of DI material shall be coated with corrosion resistant mateial such as Rilsan (Nylon 11, etc.), Halar ECTFE, etc. where necessary for the intended specific applications.
6. Butterfly valves' liner material alternatives to EPDM shall be proposed where appropriate to suit any special requirements based on fluid properties
7. For above 16 Bar applications, double eccentric butterfly valves shall be used.
8. High pressure generally refers to applications with pressure that is >= 10 Bar g.
9. Low pressure generally refers to applications with pressure that is < 10 Bar g.
10. It is noted that differing alternative materials of construction may be applicable for valve components in terms of strength, durability, resistance corrosion and compatibility with the concerned fluids for the applicable range of operating process parameters. Such materials shall be fully guaranteed for all the required characteristics and properties to ensure trouble-free operation of the valves.
11. All Ductile Iron, Cast Iron and Carbon Steel valves shall have a suitable corrosion resistant internal coating.